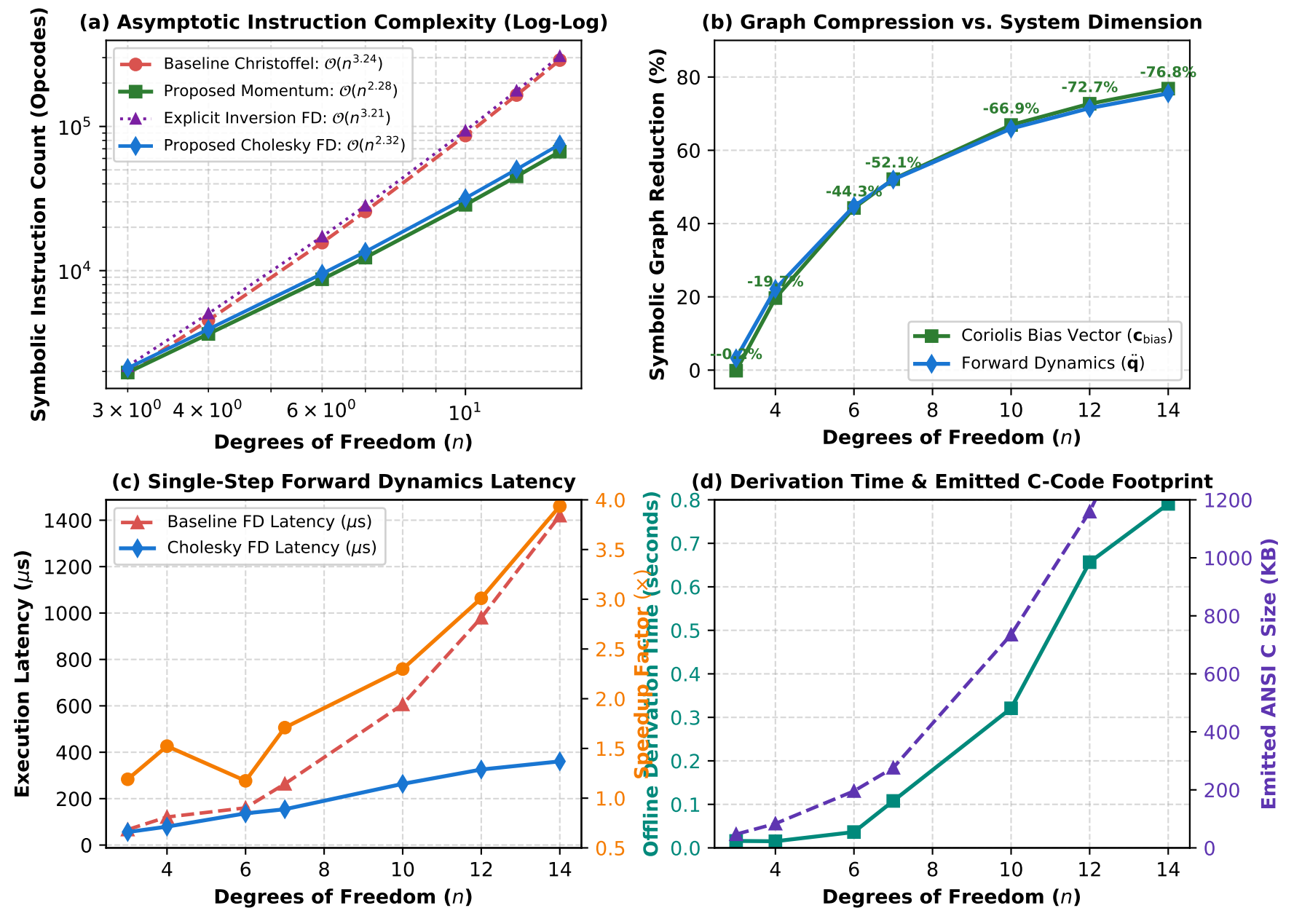
(a) Asymptotic Instruction Complexity (Log-Log)
Symbolic Instruction Count (Opcodes)
Baseline Christoffel: $\mathcal{O}(n^{3.24})$
Proposed Momentum: $\mathcal{O}(n^{2.28})$
Explicit Inversion FD: $\mathcal{O}(n^{3.21})$
Proposed Cholesky FD: $\mathcal{O}(n^{2.32})$
10^5
10^4
3×10^0
4×10^0
6×10^0
10^1
Degrees of Freedom (n)
(b) Graph Compression vs. System Dimension
Symbolic Graph Reduction (%)
-0.0%
-19.1%
-44.3%
-52.1%
-66.9%
-72.7%
-76.8%
Coriolis Bias Vector ($\mathbf{c}_{\text{bias}}$)
Forward Dynamics ($\ddot{\mathbf{q}}$)
Degrees of Freedom (n)
(c) Single-Step Forward Dynamics Latency
Execution Latency (μs)
Baseline FD Latency (μs)
Cholesky FD Latency (μs)
Speedup Factor (×)
Degrees of Freedom (n)
(d) Derivation Time & Emitted C-Code Footprint
Offline Derivation Time (seconds)
Emitted ANSI C Size (KB)
Degrees of Freedom (n)